Assessing Behavioral Responses of Canada Geese to Predator and Same-Species Vocalizations

[redacted]

Introduction

- Animals rely on sound to detect threats and coordinate group behavior.
- Canada Geese (*Branta canadensis*) use calls to communicate within flocks.[1]
- Predator calls, like those of hawks, often trigger freezing or fleeing.[2]
- Goose calls may lead to social behaviors such as vocalizing or regrouping.[3]
- We tested goose responses to three sound types: no sound, goose calls, and hawk calls.
- We expected hawk calls to cause more freezing and retreating, while goose calls would lead to more vocalizing and approaching behaviors.

AIMS

- **RQ:** Do Canada Geese exhibit different behavioral responses to hawk calls, goose vocalizations, and no sound?
- **H1:** Canada Geese will show distinct behavioral responses to different auditory stimuli: hawk calls will increase freezing and movement away (anti-predator responses), goose vocalizations will increase vocalizing and movement toward the sound (social responses), and no sound will cause minimal behavioral change.

Methods and Materials

- We observed free-living Canada Geese at Campus Pond (Amherst, MA) during daylight hours in April.
- We conducted 18 trials total (6 each of: no sound, goose call, hawk call).
- Sounds were played via Bluetooth speaker placed on the ground
- Each trial lasted 3-4 minutes while observers recorded data
- Behaviors tracked included:
    - ➔ **Freezing:** Still with neck extended
    - ➔ **Vocalizing:** Honking or calling
    - ➔ **Movement:** 2+ body lengths toward/away from sound
    - ➔ **Aggression:** Hissing, wing-flapping, chasing
- Group size was recorded at the start of each trial.
- Each trial was treated as a unit of replication. Trends were analyzed by behavior frequency across sound types.

Fig 1. Canada Geese at Campus Pond, Amherst, MA

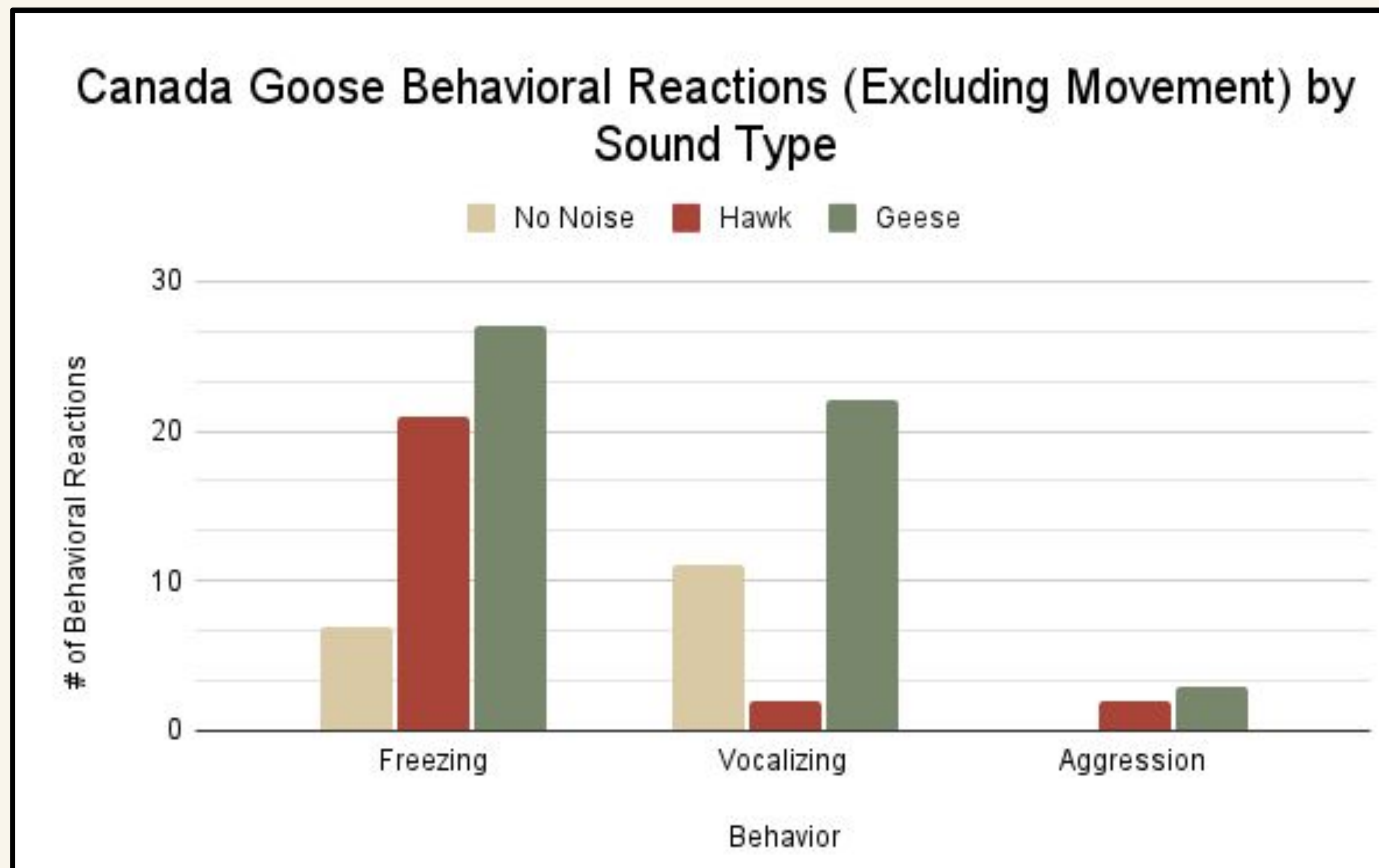

Fig 2. Behavior counts recorded during 18 trials (6 per sound)

Scan for Hawk Call

Scan for Geese Call

Thank you Seanne Clemente for your guidance throughout this project!

Results

- We used Kruskal-Wallis tests (H tests) to compare geese behaviors across the three sound conditions (hawk, goose, and no sound). The test statistic, $H(2, N = 18)$, compares ranked values across 3 groups (degrees of freedom = 2) based on 18 trials
- Freezing **differed significantly**, $H = 8.34$, $p = .015$.
- It was more common in hawk and goose trials than in no-sound trials.
- Vocalizing showed **no significant difference**, H = 5.47, $p = .065$, though it occurred most during goose trials and rarely during hawk trials.
- Movement **differed significantly**, $H = 10.50$, $p = .005$.
- Geese moved away from hawk calls, toward goose calls, and remained still during no-sound trials.

Discussion

- Our results support the hypothesis. Geese responded to hawk calls with freezing and retreat behaviors, consistent with anti-predator behavior. Goose vocalizations appeared to prompt more vocal and social behaviors, including calling back and approaching the speaker. Some freezing during goose calls may reflect social alertness rather than fear.
- These findings were limited by flock size and uncontrollable factors such as human activity and weather. Future studies could examine whether male and female geese respond differently to goose calls, as males appeared more responsive, though this could not be confirmed visually.

References

1. Canada goose sounds, all about birds, Cornell Lab of Ornithology. , All About Birds, Cornell Lab of Ornithology. (n.d.). https://www.allaboutbirds.org/guide/Canada_Goose/sounds
Policht, R., Kowalczyk, A., Łukaszewicz, E., & Hart, V. (2020). Hissing of geese: caller identity encoded in a non-vocal acoustic signal. *PeerJ*, *8*, e10197. https://doi.org/10.7717/peerj.10197
2. Wang, S., Zhou, L., Cai, J., Jiang, B., & Xu, W. (2022). Behavioral Response of Bean Goose (*Anser fabalis*) to Simulated Ship Noises at Lake. *Animals : an open access journal from MDPI*, *12*(4), 465. https://doi.org/10.3390/ani12040465
3. Guggenberger, M., Adreani, N. M., Foerster, K., & Kleindorfer, S. (2022, March 9). Vocal recognition of distance calls in a group-living basal bird: the greylag goose, Anser anser. Animal Behaviour. https://www.sciencedirect.com/science/article/pii/S0003347222000045